

David Osorio

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EDUCATION

Wesleyan University <i>Bachelor of Arts in Computer Science and Mathematics (GPA: 3.71 / 4.0)</i> – Relevant Coursework: Discrete Mathematics, Applied Data Analysis, Computer Science I/II	Middletown, CT May 2028
Houston City College <i>Associate of Science in Computer Science (GPA: 3.97 / 4.0)</i> – Graduated with Highest Honors Honors College Graduate – Relevant Coursework: Data Structures and Algorithms, Computer Organization and Architecture	Houston, TX May 2025

RESEARCH EXPERIENCE

NSF Undergraduate Researcher <i>University of Houston-Downtown</i> – Developed a Python script using pandas and scikit-learn to analyze Differential Privacy (ϵ) vs. model utility. – Integrated UCI machine learning datasets to analyze both LDP and CDP models. – Benchmarked private regressions against baseline models to model accuracy retention.	Houston, TX June 2026 – Present
Student Researcher <i>Wesleyan University</i> – Conducted quantitative research on a national U.S. survey ($N = 929$) to analyze demographic and regional predictors of climate change beliefs. – Executed bivariate and multivariate analyses (Chi-square, ANOVA, logistic/linear regression) using R.	Middletown, CT Jan. 2026

PROJECTS

NYC Crime Explorer <i>Python, pandas, scipy, Frontend Web Maps</i> – Developed an interactive web map analyzing NYPD criminal complaints and U.S. Census ACS data, color-coded by offense severity. – Built dynamic frontend filters and a backend Python script (pandas, scipy) to run regression analyses against demographic controls.	Apr. 2026
Stock Prediction Agent <i>Python, Yahoo Finance API, pandas, Matplotlib</i> – Designed an automated quantitative trading agent in Python using the Yahoo Finance API to model and forecast equity price movements. – Built a robust backtesting framework to simulate historical strategies and evaluate model performance. – Engineered a real-time module to generate trading insights and optimize risk based on historical volatility.	Mar. 2026

WORK EXPERIENCE

Service Desk Student Worker <i>Wesleyan University</i> – Resolved technical and access issues for faculty and alumni regarding Duo, Outlook, and Gmail. – Diagnosed hardware and software problems for students to restore productivity across campus systems and printers.	Middletown, CT Sept. 2025 – Present
IT Technician <i>University of Houston at Katy</i> – Troubleshoot lecture hall hardware, software, and audio-visual systems for faculty and students. – Maintained critical campus infrastructure, including library computing networks and printing systems.	Houston, TX Oct. 2024 – Aug. 2025

HONORS & AWARDS

Citation of Citizenship in the Department of Mathematics and Computer Science	May 2026
ASA DataFest CT 2026 Category Winner	Apr. 2026
Ronald E. McNair Scholar	Nov. 2025
Phi Theta Kappa Membership	Feb. 2025
Houston City College Honors College Entry/Graduation	July 2024
AP Scholar with Distinction	July 2024
Texas First Early High School Graduation Scholar	May 2024

TECHNICAL SKILLS & ADDITIONAL

Technical Skills: R, RStudio, C, C++, Python, Windows Server, Network Management, Access Control Lists
Certifications & Training: CompTIA Network+ CE, Rice University Environmental Data Academy
Languages: English (Fluent), Spanish (Fluent)